

Assignment 3: Clustering Algorithm Self-Study

Deadline: Friday, June 27th 2025

1. Algorithm Overview (20%)

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a clustering algorithm that groups together data points that are closely packed together and labels isolated points as noise.

Concepts

1. **eps** - Radius around a point to consider neighbors
2. **min_samples** - Minimum number of points needed to form a dense region
3. **Core points** - Points with $\geq \text{min_samples}$ in their eps-neighborhood
4. **Border points** - Points near core points but with fewer neighbors
5. **Noise points** - Points that are neither core nor border (outliers)

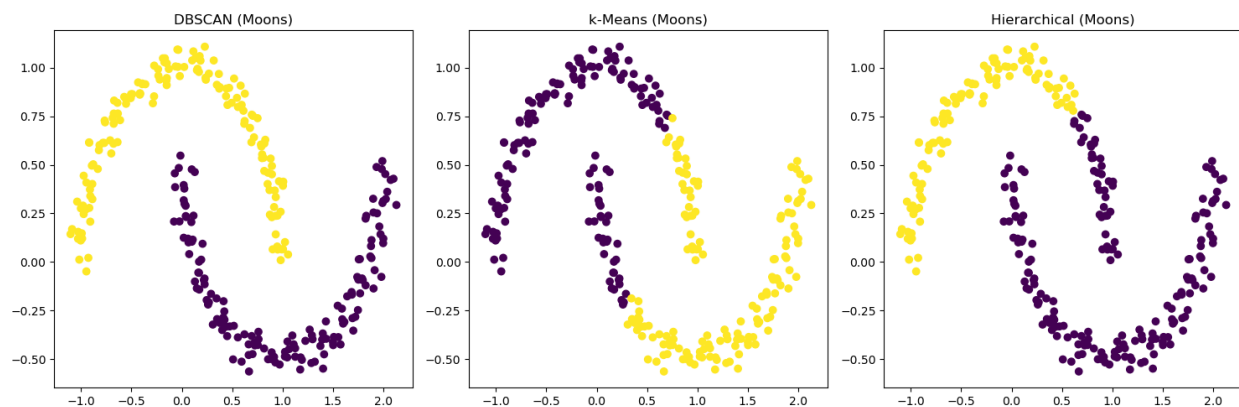
Strengths

- Handles arbitrary cluster shapes
- Handles outliers (noise) naturally

Limitations

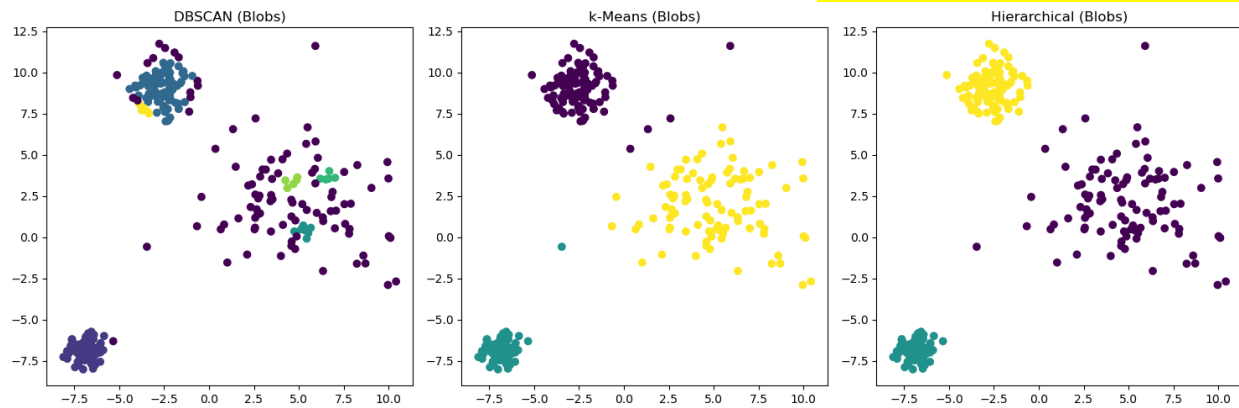
- It is slower when handling larger datasets
- Sensitive to eps and min_samples tuning.
- Clusters of varying densities can be complex to work on

2. ALGORITHM COMPARISON



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- **DBSCAN Wins on Moons:**
 - Detects non-spherical clusters; k-Means fails.
 - Labels noise points
- **DBSCAN Fails on Blobs:**
 - Struggles with varying densities (e.g., merges clusters).
 - Requires careful eps tuning.

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3. Table Update (20%)

Compare and contrast characteristics for all three algorithms:

Feature	k-Means	Hierarchical Clustering	DBSCAN
Definition	Partitioning algorithm that assigns points to k clusters based on centroids	Builds a hierarchy of clusters using distance metrics	Density-based algorithm that groups points in high-density regions and labels outliers as noise.
Approach	Iteratively minimizes variance within k clusters	Agglomerative (bottom-up) or divisive (top-down)	Starts with core points, expands clusters via reachable neighbors.
Number of Clusters	Requires predefined k	Can be determined from dendrogram but subjective	Derived from data density (no need for k).
Cluster Shape	Prefers spherical clusters	Works well with various shapes but can be unstable	Handles arbitrary shapes like moon and circles
Initialization	Randomly selects k initial centroids	No initialization needed	No initialization: Starts with core points
Result	Hard assignments—each point belongs to a single cluster	Hierarchical structure (tree/dendrogram)	Labels points as core, border, or noise (-1).
Interpretability	Moderate—cluster assignments but no hierarchy	High—dendrogram can be analyzed	Moderate: Clear clusters/noise but no hierarchy.
Strengths	Simple, fast and efficient on large datasets	Can capture hierarchical relationships	Noise-resistant, handles complex shapes, no k needed

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Limitations	Sensitive to initial centroids and k choice	Computationally expensive for large datasets	Parameter-sensitive: Struggles with varying densities
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4. Code Documentation & Submission Quality (20%)

<https://github.com/Belinda2885/BEL-BINF5507.git>